

Introduction

MSCF Investments
Fall 2021 Mini 1

- ❑ Investment management
- ❑ Learning objectives
- ❑ Big ideas
- ❑ Data sources
- ❑ Course mechanics

Duane Seppi Bio

I am the BNY Mellon Professor of Finance at the Tepper School of Business. I have also been a visiting fellow at the SEC, the University of Vienna, and Nanyang Technical University. I received my Ph.D. from the Chicago Booth Graduate School of Business.

My research interests include options, commodities, and market microstructure issues such as price manipulation, limit order markets, and market liquidity.

My research has been published in the *Review of Financial Studies*, *Journal of Finance*, *Journal of Financial Economics* and other leading finance and economics journals. I have been on the editorial boards of the *Journal of Finance*, the *Review of Financial Studies*, the *Journal of Financial Markets*, and the *Review of Finance*. I am the co-author (with N. Secomandi) of *Real Options and Merchant Operations of Energy and Other Commodities*.

Before COVID, I liked to ski, read, go to concerts, and play guitar & sing in a band.

You can see two songs I wrote and sing at:

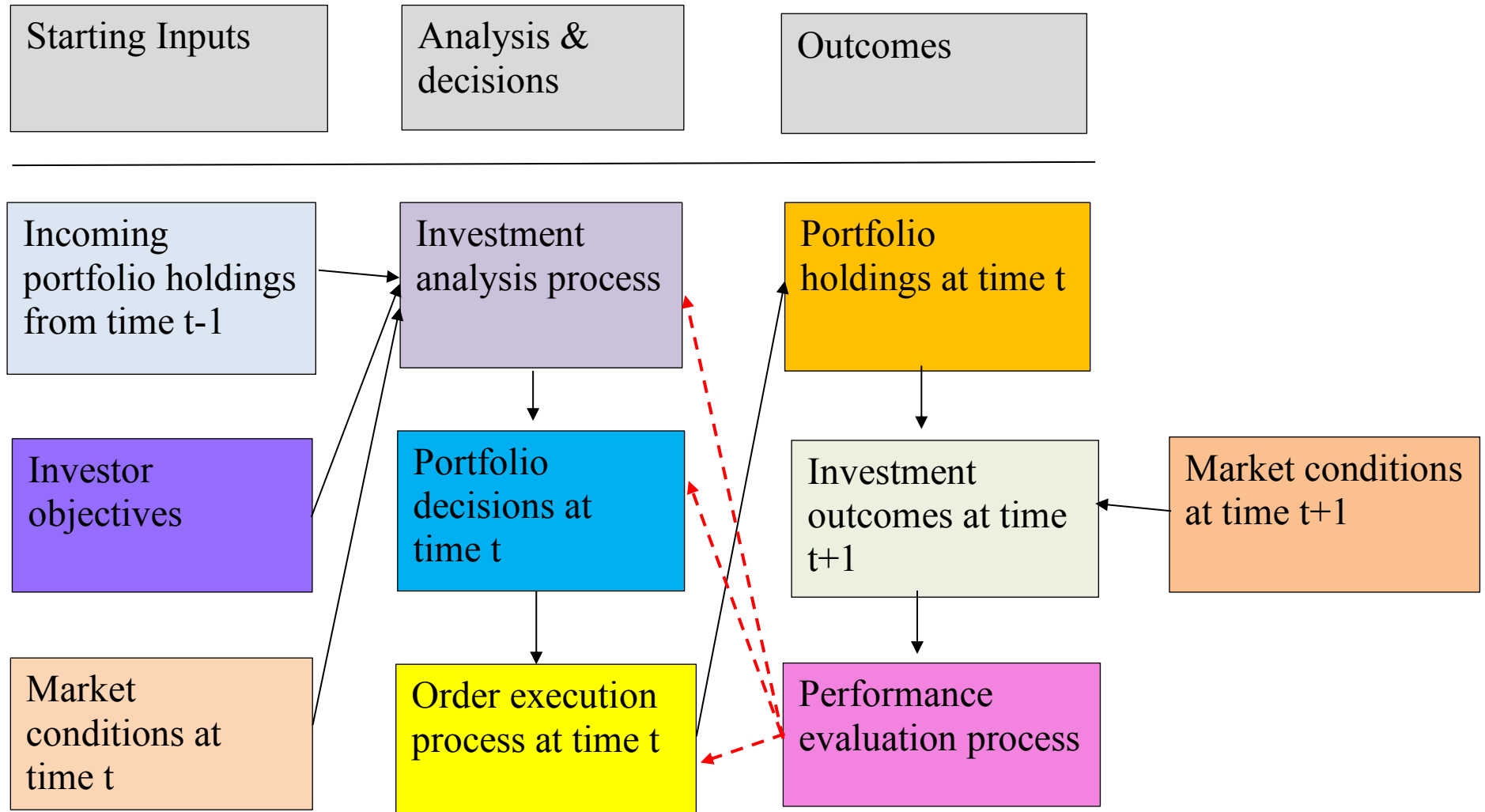
<https://www.youtube.com/watch?v=6TkOgqe5rsk>

<https://www.facebook.com/palisadeplayhouse/videos/365732260555075/>

Investment management

- Science & art of combining traded securities to create **optimal portfolios** given investor preferences, investment opportunities, tax status, and regulation.
 - An optimal portfolio is **not** the same as “**beating the market**”
- Asset management = conceptual thinking + data analysis + optimization + marketing

Investment management process



Learning objectives (1)

Able to **structure portfolio strategies** using quantitative methods

- ❑ Estimation of strategy inputs
 - Variances
 - Covariances: Linear factor models
 - Expected returns: Fama-MacBeth regressions, predictive models
- ❑ Portfolio formation
 - Mean-variance optimization
 - Structuring using linear factor models
 - Trading on anomalies
- ❑ Order execution
 - Market architecture: Limit order markets, over-the-counter (OTC) dealer markets
 - Regulation
 - High-frequency trading
- ❑ Performance evaluation
 - Benchmarking using Fama-French factors and extensions
 - Estimate and interpret Jensen's alpha, Sharpe Ratios, and "style analysis"

Learning objectives (2)

Make decisions guided by an understanding of **fundamental asset-pricing concepts**

- ❑ Models of asset pricing and expected returns
 - Campbell-Shiller
 - Concepts of priced risk and unpriced randomness
 - Equilibrium risk/reward trade-off
 - CAPM and APT
 - Role of sentiment
- ❑ Structure & sources of return variability and co-movement
 - Linear factor models
 - Public news, private information, sentiment, demand imbalances and trading noise
- ❑ Information and price discovery
 - Efficient markets
 - Sentiment
 - Behavior finance

Learning objectives (3)

Use empirical evidence from **data analysis** to guide asset management

- ❑ Interpret statistical evidence
 - OLS regression, t-statistics, p-values, R^2
 - Reading cutting-edge empirical research: Some classics but also lots of current research
 - Event studies, linear regression models, Fama-MacBeth, Campbell-Shiller decompositions
- ❑ Understand current empirical evidence guiding current asset management strategies
 - Properties of important financial factors: Fama-French, momentum, macro, political
 - Anomalies and asset-pricing puzzles: What are they? Are they real?
 - Properties of different types of information: Numerical, text, sentiment proxies
- ❑ Hands-on practice using Bloomberg as a source of data
- ❑ Original empirical project

BIG Ideas (1)

How to think about traded securities:

Traded securities are **bundles** of cash-flows (CF) & other characteristics:

- Different securities have ...
 - Different **risk exposures** to different types of CF randomness **inherited** from securities' differing claims on corporate earnings, gov't finances, borrower income, etc.
 - Different **other characteristics**: CF timing (i.e., time-value-of-money), firm size, index membership, industry, growth/value, country, credit rating, ESG, ¹ etc.
 - Different **expected returns**
- Individual securities are “**building blocks**”
- Investors structure **portfolios** to combine individual stocks, bonds, etc. to construct **preferred combinations** of risk, other characteristics, and expected return

¹ Environmental, Social, Governance
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BIG Ideas (2)

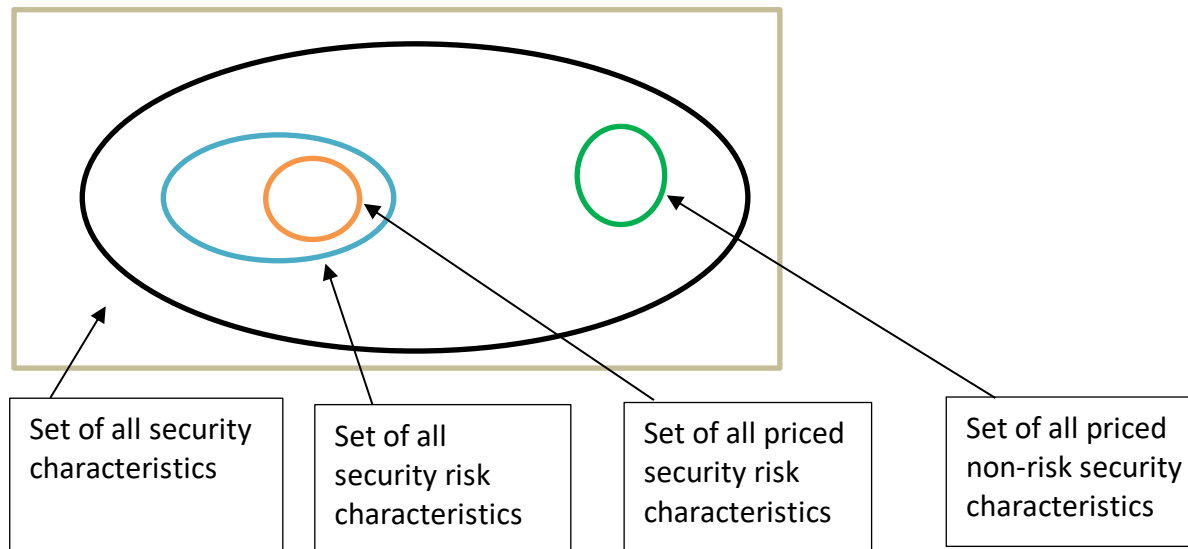
Asset pricing & where investment properties in traded assets come from

- ❑ Opportunity set of available bundles of portfolio risk, other characteristics & $E[\text{ret}]$ induced by ...
 - The **supply** of risks and other characteristics given individual securities from real economy.
 - How market “**prices**” risk and characteristics in individual securities
- ❑ Not all combinations of risk, other characteristics, and $E[\text{ret}]$ are feasible in market.
- ❑ Key ideas from economics:
 - The market **prices** individual traded securities.
 - Investors **structure** portfolios to hold using traded securities.
 - How market prices individual securities is **endogenous**: Depends on aggregate demand for individual assets given types of portfolios investors want to form using those assets.
 - Investor demand: $D(p \mid \text{statistical risk, other characters, investor preference})$
- ❑ Equilibrium:
 - Market-clearing and pricing: Demand for traded securities = Supply of traded securities
 - Demand for traded risks and characteristics = Supply of traded risks and characteristics
 - In well-functioning market, risks and characteristics should be priced **consistently** across different securities. Impact on pricing should not depend on security ID. Otherwise trading opportunity.

BIG Ideas (3)

Priced risks, unpriced randomness, and other priced and unpriced security characteristics

- ❑ Risks & characteristics are **priced** if cause asset prices to differ from $\Sigma E[\text{future cash flows}]$
 - Time value of money
 - Risk-reward trade-off
 - But what is risk?
 - Priced risk $\neq$ randomness per se
 - Some randomness is priced, but not all. Which randomness is priced?
 - How to measure risk?
 - Examples?



BIG Ideas (4)

Two views about information & how to earn high returns:

- Efficient Market Hypothesis
 - Prices set correctly given available information & preferences.
 - Earn high $E[\text{ret}]$ as **compensation** for bearing priced risk & holding other unattractive characteristics
 - Learning dynamics: Market learns over time & corrects past pricing errors.
- Behavioral finance
 - Two versions: Irrational and rational
 - Cognitive biases in investor decision-making lead to market pricing errors
 - Earn high $E[\text{ret}]$ by **trading mispriced securities** and waiting for short-term corrections
 - Sentiment and rational asset pricing
 - Asset pricing depends on what investors care about, and that can change over time
 - Earn high $E[\text{ret}]$ by **accommodating** other investors' sentiment demands

BIG Ideas (5)

- Prices are a **social construct** involving humans with computational aids
 - Prices are **not pure statistical forecasts**. Also depend on investor preferences, biases, & decisions → Aggregate demand for CF and other characteristics
 - Prices are solutions to a **market-clearing** problem given asset demands of **intelligent optimizing** investors with **cognitive limitations** and **biases**
 - Market beliefs: What probabilistic beliefs are **priced into** market-clearing prices?
 - Consensus market beliefs may be different from probabilistic “**views**” guiding individual investor decisions (e.g., a hedge fund that does ML to predict returns better than market).
 - Market beliefs can differ from best **statistical forecasts**. Can be for both rational reasons (preferences) or errors.

What this course is and isn't

- “Quant finance” once meant option pricing & term structure modeling. Now also quant investing
 - “p-measure” quants rather than “q-measure” quants
 - Everything you need to know about quant finance aside from derivative pricing
 - Even if you want a career in options, helps to know what drives underlying asset dynamics
- Hybrid investments/asset pricing class
 - Not a standard “textbook” investments class
 - Based on asset pricing theory rather than one-off formula punchlines
 - Emphasis on data analysis & on interpreting empirical evidence of your own and others
 - Will look at recent research and current topics
- Gateway course into MSCF asset management curriculum stream

Online data sources (in addition to Bloomberg)

- ❑ Macroeconomic data
 - St. Louis Federal Reserve (FRED) website
https://fred.stlouisfed.org/?gclid=EAlaIQobChMIkl2m-7LB6wIVMvC1Ch3FhAGnEAAYASAAEgJiivD_BwE
 - NBER business cycle data
<https://www.nber.org/cycles.html>
- ❑ Fama-French factors
 - Ken French's website:
https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
- ❑ Aggregate market dividend, earnings, and return data
 - Robert Shiller's website:
<http://www.econ.yale.edu/~shiller/data.htm>
- ❑ Political risk
 - Economic policy Uncertainty
https://www.policyuncertainty.com/us_monthly.html

Course Mechanics

Weekly homework

- ❑ Individually or work on a team with one other person
- ❑ OK to talk to others in the class, but no sharing of programs or written answers
- ❑ 10 points per assignment. HW graded on “effort”

Quiz

- ❑ In week 5 based on material from weeks 1-4. No make-up quizzes.
- ❑ 20 minutes
- ❑ 10 points

Project

- ❑ Regression analysis: Ex ante prediction or ex post attribution for returns or risk
- ❑ Due before last class
- ❑ Team of 2
- ❑ 50 points

Final exam

- ❑ Cumulative covering all topics in the course
- ❑ 100 points

Office hours

- ❑ Schedule TBD, also after class, or by appointment